

Full Length Article

Optimum primary equivalence ratio for rich-lean two-stage combustion of non-premixed ammonia/methane/air and ammonia/hydrogen/air flames in a swirling flow

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ABSTRACT

Rich-lean two-stage combustion concept was proposed for reducing emissions from ammonia-fueled gas turbine combustors. The primary combustion zone should be maintained at an optimum primary equivalence ratio for the simultaneous reduction of NO and unburned NH₃; however, the optimum primary equivalence ratio has been reported only for a limited number of fuel blends and ammonia mixing ratios. This study experimentally and numerically investigated the optimum primary equivalence ratio and studied its application in the rich-lean two-stage combustion of ammonia/methane/air and ammonia/hydrogen/air flames over a wide range of ammonia mixing ratios under atmospheric pressure. This study identifies factors determining the optimum primary equivalence ratio. The emissions were measured using a swirl combustor for single- and two-stage combustion. Unstretched freely propagating premixed flame simulations were performed using CHEMKIN-PRO for investigating the effects of chemical reactions on the optimum primary equivalence ratio. In both single- and two-stage combustion, the optimum primary equivalence ratios of non-premixed ammonia/methane/air and ammonia/hydrogen/air flames decreased linearly and exponentially with increasing ammonia mixing ratios, respectively. The rich-lean two-stage combustion eliminated unburned NH₃ and minimized NO over a wide range of ammonia mixing ratios for non-premixed ammonia/methane/air and ammonia/hydrogen/air flames. The numerical results confirm that OH radical plays an important role in determining the optimum primary equivalence ratio.

1. Introduction

Direct ammonia combustion is a promising approach for decarbonizing power generation and the industrial and transportation sectors [1–5]. Ammonia is considered a promising hydrogen energy carrier for transporting renewable energy. Ammonia liquefies at 8.5 atm and 293 K, and it has a high volumetric hydrogen density of 121 kg-H₂/m³ [2]. Therefore, ammonia is easier to handle than hydrogen and the storage and transportation costs of ammonia over long distances are lower than those of hydrogen [6]. In addition, the cross-ministerial Strategic

Innovation Promotion program (SIP) “Energy Carriers” project of the Japanese government proved the effectiveness of ammonia as a fuel [7]. The development of the ammonia fuel industry was specified in the Japanese policy, “Green Growth Strategy Through Achieving Carbon Neutrality in 2050” [8]. Ammonia is a carbon-free fuel that does not emit CO₂ during combustion. Blue and green ammonia are produced from fossil fuels because of the CO₂ captured during production, and the renewable energy sources have zero CO₂ emissions in all processes including production, respectively. Blue and green ammonia production plants are being planned worldwide, and therefore, there is an urgent

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need for the development of science and technology for ammonia combustion.

Challenges associated with ammonia combustion include its low flammability and poor emission characteristics [2]. The laminar burning velocity of premixed ammonia/air flame is one-fifth of that of premixed methane/air flame, and its adiabatic flame temperature is about 200 K lower than that of premixed methane/air flame. The range of the flammable equivalence ratios was narrow (0.63–1.4), which implies that the volume of ammonia-fueled combustors can be larger, and that the operating conditions can be more restricted in gas turbines. Further, ammonia combustion emits Fuel- NO_x and unburned NH_3 depending on combustion conditions. Under most conditions, the main component of Fuel- NO_x is NO, which is a major air pollutant. N_2O , which is produced in the low-temperature region of the combustor, should be considered because the global warming potential (defined as 1 for the warming effect of CO_2) is 273 [9]. Unburned NH_3 is classified as an extremely hazardous substance in many countries, and it must not be emitted from the combustor because high levels of ammonia can irritate and burn the skin, mouth, throat, lungs, and eyes. Thus far, various studies have attempted to address these challenges, as described below.

This study focused on direct ammonia combustion in gas turbines. Ammonia-fueled gas turbines were considered in the 1960 s; however, their low flammability prevented their commercialization [10,11]. Recently, Hayakawa et al. [12] successfully stabilized gaseous pure ammonia/air flames in a swirl combustor and reported product gas characteristics. This research was extended to demonstrate power generation by gaseous pure ammonia combustion in a 50 kW micro gas turbine using green ammonia at the Fukushima Renewable Energy Institute (FREA) of the National Institute of Advanced Industrial Science and Technology (AIST) [13]. In addition, liquid ammonia spray combustion was proposed to help ammonia-fueled gas turbines generate higher power, improve startability and load followability, and reduce capital and operating costs [14,15].

Co-firing with other fuels is used as a method for improving the flammability of ammonia/air flames. The addition of methane [16], hydrogen [17], n-heptane, isooctane [18], methylcyclohexane, and toluene [19] to the ammonia/air flames increases the laminar burning velocity. Khateeb et al. [20–22] showed that the addition of methane or hydrogen to ammonia/air flames in a swirl combustor expanded the flame stability limits. The addition of methane and hydrogen increased the maximum NO emissions and caused flashbacks in the premixed flames. Nitrogen dilution [21], steam addition, [23] and high-pressure conditions [24] have been proposed as countermeasures for these problems. Mashruk et al. [25] recommended a 20/55/25 vol% ternary blend of methane/ammonia/hydrogen/air flames caused by their low overall emissions and high stability. Chai et al. [26] summarized the chemical kinetics of ammonia/methane/air and ammonia/hydrogen/air flames in a review paper. A common feature of methane and hydrogen addition is that they increase the O/H radical pool in ammonia/air flames. In addition to co-firing with other fuels, an increase in the temperature of the mixture effectively enhances ammonia/air flames. Kanoshima et al. [27] reported the temperature and pressure exponents of the laminar burning velocity for ammonia/air flames. The temperature exponents of ammonia are larger than those of methane. Takeishi et al. [28] observed that the laminar burning velocity and flame temperature increased with an increase in the oxygen fraction of the mixture. Tang et al. [29] and Lin et al. [30] demonstrated the effectiveness of a support using a gliding arc plasma for ammonia/air flames.

Rich-lean two-stage combustion concept was proposed to control emissions from ammonia flames [31]. Kurata et al. [32] operated an ammonia-fueled micro-gas turbine using this method to generate electricity and demonstrated low- NO_x emissions from the combustor. Moreover, Ito et al. [33] achieved low- NO_x emissions and a high combustion efficiency of 99.6% simultaneously in ammonia/methane co-firing using an existing 2 MW gas turbine combustor with minimal modifications.

Rich-lean two-stage combustion simultaneously reduces NO and unburned NH_3 based on the trade-off relationship between NO and unburned NH_3 emissions from ammonia flames. The primary combustion zone burned under rich conditions that minimized both NO and unburned NH_3 , whereas the secondary combustion zone burned under ultra-lean conditions. Unburned NH_3 is burned with the hydrogen emitted from the primary combustion zone and with the secondary air from the secondary combustion zone. The performance of the rich-lean two-stage combustion depends on the optimum primary equivalence ratio ϕ_{opt} for the simultaneous reduction of NO and unburned NH_3 in the primary combustion zone. The method for obtaining ϕ_{opt} from the experiments in this study is discussed below in Section 3.5. Previous studies using swirl combustors reported ϕ_{opt} for gaseous pure ammonia/air flames [12,34–38], gaseous ammonia/methane/air flames with an ammonia heat fraction E_{NH_3} up to 0.3 [39], and liquid ammonia/methane/air flames with $E_{\text{NH}_3} = 0.7$ [14]. Further, previous studies using stagnation burners reported ϕ_{opt} for pure gaseous ammonia/air flames [40] and gaseous ammonia/methane/air flames with E_{NH_3} up to 0.6 [41]. Okafor et al. [39] and Kovaleva et al. [41] showed that ϕ_{opt} shifted towards the lean side with an increase in E_{NH_3} concentration. However, ϕ_{opt} is known only under limited fuel blends and E_{NH_3} conditions; the mechanism of the shift to the lean side with an increase in E_{NH_3} is yet to be clarified. These findings are essential for promoting direct ammonia combustion in gas turbines and replacing existing gas turbine combustor fuels with ammonia in a phased manner.

In this study, ϕ_{opt} for the simultaneous reduction of NO and unburned NH_3 in ammonia/methane/air and ammonia/hydrogen/air flames was investigated experimentally using a swirl combustor over a wide range of $E_{\text{NH}_3} = 0.10$ –1.00 under atmospheric pressure. Rich-lean two-stage combustion at atmospheric pressure was conducted for verifying the control of NO and unburned NH_3 . Further, factors for determining ϕ_{opt} were investigated using 1D unstretched laminar flame simulations.

2. Apparatus and procedures

2.1. Experimental setup

Fig. 1 shows a schematic of the experimental setup. The air was supplied by a compressor with a dryer, and its flow rate was controlled using a thermal mass flow controller (Brooks, SLA5853) with an accuracy of F.S. 0.2%. Ammonia (99.9%), methane (99.0%), and hydrogen (99.99%) were supplied from each cylinder, and their flow rates were controlled by needle valves and thermal mass flow meters (KOFLOC, 3760 and 3765 series) with an accuracy of F.S. 1.0–1.5%. Before the experiments, all mass flow controllers and meters were calibrated with actual gas under real conditions using a standard flow meter traceable to

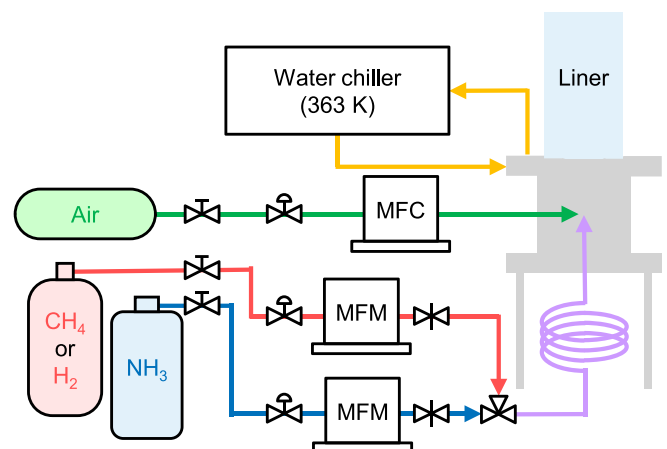


Fig. 1. Schematic of the experimental setup.

the national measurement standards. An ammonia cylinder was maintained at ~ 313 K to prevent the upstream pressure from decreasing. The fuels (ammonia, methane, or hydrogen) were premixed in a fuel-mixing zone of approximately 1 m. A chiller (Kawata, TWF-75LDA) circulated the cooling water at ~ 363 K for preventing the burnout of the combustor base plate.

Experimental parameters such as flow rate and temperature were recorded using a data logger (HIOKI, LR8450), and the equivalence and ammonia mixing ratios were calculated in real time. The ammonia mixing ratio defined by the ammonia heat fraction E_{NH_3} and co-firing fuel volume fraction x_i ($i = \text{CH}_4$ or H_2) of the fuel are respectively expressed as

$$E_{\text{NH}_3} = \frac{Q_{\text{NH}_3} H_{\text{L, NH}_3}}{Q_i H_{\text{L, } i} + Q_{\text{NH}_3} H_{\text{L, NH}_3}}, \quad (1)$$

$$x_i = \frac{Q_i}{Q_i + Q_{\text{NH}_3}}, \quad (2)$$

where Q_{NH_3} and Q_i represent the flow rates [SLM (Standard liter per minute)] of ammonia and the co-firing fuel, respectively. In this study, the standard condition is defined as 1 atm at 293 K. Further, $H_{\text{L, } i}$ represents the lower heating value of each fuel ($H_{\text{L, NH}_3} = 316.84$ kJ/mol, $H_{\text{L, CH}_4} = 802.30$ kJ/mol, and $H_{\text{L, H}_2} = 242.32$ kJ/mol).

2.2. Swirl combustor

Fig. 2 shows a schematic of the base part of the swirl combustor cross section. The combustor was designed for liquid ammonia spray combustion, and its geometry was reported by Okafor et al. [14]. In this study, a dummy nozzle without holes was installed instead of a spray nozzle, and the combustor was modified to a non-premixed combustor for gaseous ammonia combustion. Air was introduced horizontally through the tangential swirler and mixed well with the fuel injected vertically through the 16 fuel injection holes around the dummy nozzle. Because of the use non-premixed combustor, pure hydrogen/air flames can be stabilized without flashbacks. The combustor base plate was protected from burnout by cooling the water to ~ 363 K.

The tangential swirler enhanced the swirling flow to a greater extent than the axial swirler used in our previous studies on gaseous ammonia combustors [31,37]. The swirl number was calculated using

$$S = \frac{\pi D^2}{4A_s}, \quad (3)$$

where D and A_s represent the inside diameter of the tangential swirler (25 mm) and total slit area of the tangential swirler (8 slits $\times 3.74 \times 7$ mm²), respectively. The swirl number was calculated as 2.34 [42]. This swirl number differs from the general swirl number proposed by Chigier et al. [42] because the air introduced through the swirler has no axial momentum.

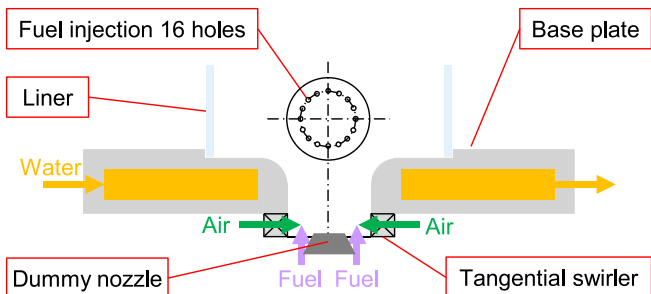


Fig. 2. Cross-sectional schematic of the base part of the swirl combustor.

2.3. Particle image velocimetry

Flow fields under the non-reacting conditions of the single-stage setup shown in Fig. 3a and the two-stage setup shown in Fig. 3b are measured using particle image velocimetry (PIV). The combustor liners are composed of a 3-mm-thick cylindrical quartz glass. A liner with an inner diameter of 72 mm and a height of 150 mm was employed for the single-stage setup. A secondary air injector [37] for the two-stage setup; a liner with an inner diameter of 72 mm and height of 42 mm were added on top of the single-stage setup. The secondary air injection holes (8 holes $\times 10 \times 10$ mm²) at a height of 158 mm were positioned at the center of the combustor.

Table 1 lists the primary PIV settings used in this study. The PIV optics consisted of a 532 nm Nd:YAG double-pulse laser (New Wave Research, Solo PIV 120), mirrors, sheet-forming optics, and a dual-shutter camera (Flowtech Research, FtrDSC) attached to a lens (Nikon, AI Nikkor 50 mm f/1.2S). The sheet-forming optics included a beam expander (SIGMAKOKI, LBED-10), planoconvex cylindrical lens ($f = 700$ mm), aperture, and planoconcave ($f = -30$ mm) and planoconvex cylindrical lenses ($f = 150$ mm). The trigger signal was generated using a digital delay generator (Flowtech Research, FtrVSD2000) and a camera control software (Flowtech Research, FtrCAM) was used for the operation setting and image acquisition. The seeding particles were olive oil mists ($1\text{--}5$ μm) generated by an oil mist generator (Flowtech Research, FtrOMG). PIVlab [43] was used for processing and analyzing the PIV images.

The flow field of the single-stage setup with a primary air flow rate, $Q_{\text{pri}} = 100$, and 300 SLM (the mean swirler inlet velocity, $U_m = 8$ and 24 m/s at the standard condition) was already measured by Okafor et al. [14]. In this study, the flow field of the single-stage setup with $Q_{\text{pri}} = 150$ SLM ($U_m = 12$ m/s at the standard condition), and the two-stage setup with primary and secondary air flow rates, Q_{sec} and $Q_{\text{pri}} = 150$ SLM, were measured. These conditions correspond to the experimental combustion conditions in this study although the effect of fuel flow is not considered.

2.4. Measurement of product gas characteristics

The product gas characteristics of the single-stage combustion shown in Fig. 3a and the two-stage combustion shown in Fig. 3b are measured using a Fourier transform infrared spectroscopy (FTIR) gas analyzer

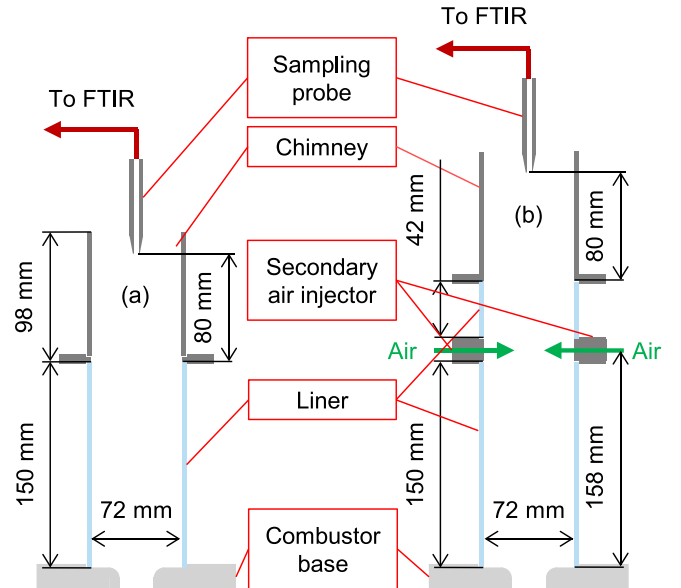


Fig. 3. Schematic of (a) single- and (b) two-stage setups of the swirl combustor.

Table 1

Main PIV settings.

Parameter	Setting
Acquisition frequency	15 Hz
Number of images	400 (200 pairs)
Pulse separation, Δt	50 μ s
PIV algorithm	FFT window deformation
Initial interrogation area	32 $\times$ 32 pixel
Final interrogation area	16 $\times$ 16 pixel
Overlap	16 pixel

(Best Instruments, BOB-2000FT). The H_2 and O_2 emissions, which cannot be measured by FTIR, were measured using a thermal conductivity H_2 analyzer and a paramagnetic (dumbbell) O_2 analyzer attached to the FTIR gas analyzer, respectively. The accuracy of these analyzers was F.S. $\pm$ 2%. The results obtained under conditions in which even one chemical species was above the measurement range of the analyzers were discarded because the accuracy of all other chemical species could not be guaranteed. The chemical species in the product gas were presented as mole fractions in units of ppm or %.

The measurement point was 80 mm downstream from the top of the liner at the center of the combustor, as in the previous study [14]. Fig. S1 shows that there is no radial variation in the product gas characteristics at the height of this measurement point. The combustion products are introduced into the sampling line of the FTIR gas analyzer through a sampling probe heated to ~ 423 K using a ribbon heater. The temperature of the sampling gas was adjusted to 465 ± 1 K in the FTIR cell. A stainless-steel chimney with a height of 98 mm at the top of the liner prevented the outside air from being sampled. Measurements were performed for 60 s at 5 Hz after all chemical species concentrations reached a steady state for each condition. The measured data reported in this study are the time-averaged wet concentrations of the species. At least two measurements were performed on different dates and times to confirm the repeatability of the experimental results.

All combustion experiments were conducted at 1 atm, 293 K. The primary and secondary air was supplied to the combustor at 293 K. Direct images of the flames were captured using a single-lens reflex camera (Nikon, D610) and a single focal-length lens (Nikon, AI Nikkor 50 mm f/1.2S). Camera settings such as the ISO and shutter speed were adjusted so that the images were close to those observed by the human eye. At the beginning of the experiments, the combustor was ignited manually from the top using a gas torch. In the single-stage combustion, the equivalence ratio in the primary combustion zone ϕ_{pri} varied by modifying the fuel flow rate (thermal input power) while the Q_{pri} was fixed. The ϕ_{opt} for the simultaneous reduction of NO and unburned NH_3 was investigated under a wide range of $E_{NH_3} = 0.10$ –1.00 ($x_{CH_4} = 0.78$ –0.00 or $x_{H_2} = 0.92$ –0.00) for gaseous ammonia/methane/air, ammonia/hydrogen/air, and pure ammonia/air flames. In the two-stage combustion, the control of emissions using the rich-lean two-stage combustion strategy was verified with ϕ_{opt} identified in the single-stage combustion experiments. The ϕ_{pri} and the overall equivalence ratio ϕ_{ovr} were respectively expressed by

$$\phi_{pri} = \frac{\alpha Q_i + 0.75 Q_{NH_3}}{0.21 Q_{pri}}, \quad (4)$$

$$\phi_{ovr} = \frac{\alpha Q_i + 0.75 Q_{NH_3}}{0.21 (Q_{pri} + Q_{sec})}, \quad (5)$$

where Q_{pri} , Q_{sec} , Q_{NH_3} , and Q_i represent flow rates [SLM] of primary air, secondary air, ammonia, and mixed fuel, respectively. For ammonia/methane/air flames, $\alpha = 2$; for ammonia/hydrogen/air flames, $\alpha = 0.5$; and for pure ammonia/air flames, $\alpha = 0$. The Q_{sec} was adjusted to keep ϕ_{ovr} constant in the two-stage combustion.

2.5. Numerical analysis

1D unstretched adiabatic freely propagating premixed laminar flame simulation was conducted using the flame speed calculation model in CHEMKIN-PRO [44] for investigating the effect of chemical kinetics on ϕ_{opt} . The pressure and unburned gas temperature were set to 1 atm and 298 K, respectively. The calculation domain size L was set to 0.1 m, and the mole fractions at the position 0.04 m downstream from the position of the maximum heat release rate were adopted as the numerical results. The mechanism proposed by Okafor et al. [16] and Han et al. [45] were used for ammonia/methane/air and ammonia/hydrogen/air flames, respectively. In the case of pure ammonia/air flame, both mechanisms were used. The integrated rate of production (ROP), $I_{R,i}$ was used to evaluate the contribution of each elementary reaction to the production and consumption of a given chemical species,

$$I_{R,i} = \int_0^L \dot{\omega}_{R,i} dx, \quad (6)$$

where $\dot{\omega}_{R,i}$ represents the ROP of the species i through the reaction R , and L represents the calculation domain.

3. Results and discussion

3.1. Flow field of the combustor

Fig. 4 shows the flow field averaged over 200 PIV images of the (a) single- and (b) two-stage setups under non-reacting conditions. The direction of the flow corresponds to the direction of the arrow, and the length of the arrow and the background color correspond to the magnitude of the velocity. Measurements were captured at different times for two areas (upper and lower 80 mm heights) in the single-stage setup and for three areas (upper and lower 80 mm heights and secondary combustion zone) in the two-stage setup. The lower part of the secondary combustion zone in the two-stage setup could not be measured because the secondary air injector blocked the laser sheet.

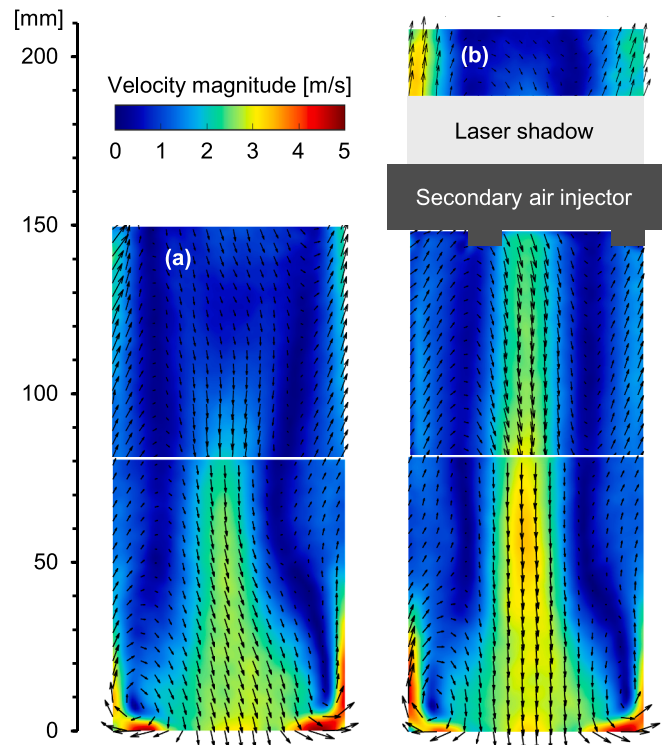


Fig. 4. Flow field of (a) single- and (b) two-stage setups under non-reacting conditions.

Fig. 4a shows the flow field of the single-stage setup at $Q_{\text{pri}} = 150$ SLM ($U_m = 12$ m/s). The mainstream flowed along the bottom and the sides of the combustor. A bell-shaped central recirculation zone characteristic of tangential swirlers was formed at the base of the combustor liner, whereas an outer recirculation, commonly observed in combustors with an axial swirler, was not formed. The central recirculation zone produced a backflow of approximately 2 m/s to a height of approximately 100 mm. The backflow at the center of the liner exit suggests that air entered from outside. This flow field was similar to that at $Q_{\text{pri}} = 100$ and 300 SLM ($U_m = 8$ and 24 m/s) measured by Okafor et al. [14].

Fig. 4b shows the flow field of the two-stage setup at $Q_{\text{pri}} = Q_{\text{sec}} = 150$ SLM corresponding to $\phi_{\text{ovt}} = 0.5$. The injection velocity of the secondary air from each hole of the secondary injector was 3.13 m/s. The secondary air increases the flow velocity near the outlet wall of the secondary combustion zone. Comparing the flow fields of the single- and two-stage setups revealed that the shape and velocity of the central recirculation zone below a height of 80 mm did not change significantly. However, the height of the central recirculation zone, where backflow occurred at ~ 2 m/s, extended to a height of 150 mm in the two-stage setup. This implies that some of the secondary air supplied by the secondary air injector entered the primary combustion zone in the two-stage setup. Although the flow of secondary air into the primary combustion zone appears to be limited under these conditions, its introduction undermines the concept of a rich-lean two-stage combustion. The height of the secondary air injection should be considered carefully when designing a combustor.

3.2. Single-stage combustion

3.2.1. Ammonia/methane/air flames

Fig. 5 shows the NO_x and unburned NH_3 emissions during the single-stage combustion of ammonia/methane/air and pure ammonia/air flames. The results for the other E_{NH_3} conditions are shown in Fig. S2. The measurable equivalence ratios were limited by the flame stability limit on the lean side, HCN measurement range (500 ppm) on the rich side, and NO measurement range (5000 ppm).

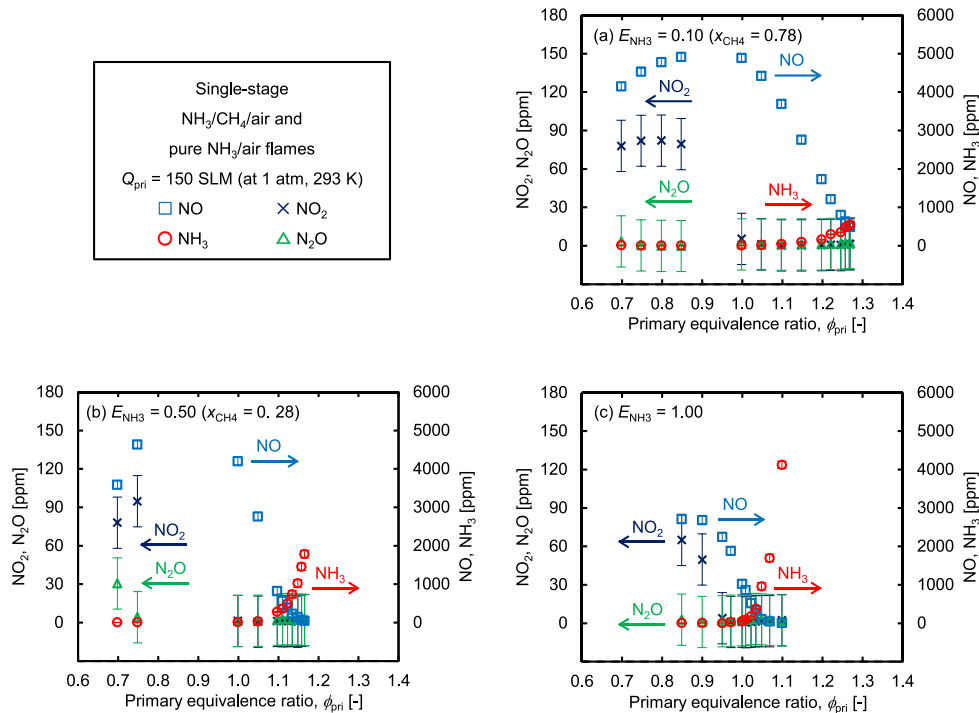


Fig. 5. NO_x and unburned NH_3 emissions in the single-stage combustion of ammonia/methane/air and pure ammonia/air flames at (a) $E_{\text{NH}_3} = 0.10$ and (b) $E_{\text{NH}_3} = 0.50$, and (c) $E_{\text{NH}_3} = 1.00$.

NO emissions peaked under the lean conditions for all E_{NH_3} conditions; however, the measurements could not be performed under certain conditions because the emissions were larger than the measurement range of the analyzer. The maximum value of NO emissions increased with increasing E_{NH_3} up to approximately $E_{\text{NH}_3} = 0.3$ and then decreased. This trend is in good agreement with the previously reported experimental results [20,41,46]. Under rich conditions, unburned NH_3 was emitted with a decrease in NO . As in the ammonia/methane/air flames [39,41], NO and unburned NH_3 exhibited a tradeoff relationship, thereby creating a low-emission window around ϕ_{opt} , where the sum of the emissions of NO and unburned NH_3 was the lowest. ϕ_{opt} was observed for all E_{NH_3} conditions and decreased with increasing E_{NH_3} , even under high E_{NH_3} conditions, which has not been studied previously. The specific values of NO and unburned NH_3 at the ϕ_{opt} are discussed in Section 3.5. An increase in E_{NH_3} narrowed the low-emission window. In other words, higher E_{NH_3} conditions have a higher sensitivity of the equivalence ratio to NO and unburned NH_3 near ϕ_{opt} , and they require more precise control of ϕ_{pri} . NO_x was dominated by almost 100% NO ; however, some lean flames emitted NO_2 and N_2O . The NO_2 emissions correlate with NO emissions, and the peak of NO_2 emissions appear to be slightly on the lean side of the NO peak. Fig. 6a shows that H_2 was emitted from $\phi_{\text{pri}} = 0.95$ –1.10, and then, it increased linearly with an increase in ϕ_{pri} . In experiments under low E_{NH_3} conditions up to $E_{\text{NH}_3} = 0.3$, Okafor et al. [39] could not confirm the effects of E_{NH_3} on H_2 emissions; however, a comparison over a wide range of E_{NH_3} conditions in this study revealed that an increase in E_{NH_3} promotes H_2 emissions under rich conditions. This increase in H_2 emissions with E_{NH_3} can be attributed to a change in the main source of H_2 from the methane chemistry to the ammonia chemistry. HCN emissions were observed under rich conditions with an increase in the ϕ_{pri} , and the emissions increased rapidly, as shown in Fig. 6b. The lowest ϕ_{pri} , where HCN emissions were observed, decreased with an increase in E_{NH_3} except $E_{\text{NH}_3} = 1.0$, shifting the HCN emissions profile towards the lean side. Further, HCN is formed in ammonia/methane/air flames through the reactions of NO or N with CH_3 , CH_2 , and CH [39], and it is consumed by O and OH radicals [47]. The reason that E_{NH_3} affected HCN at rich

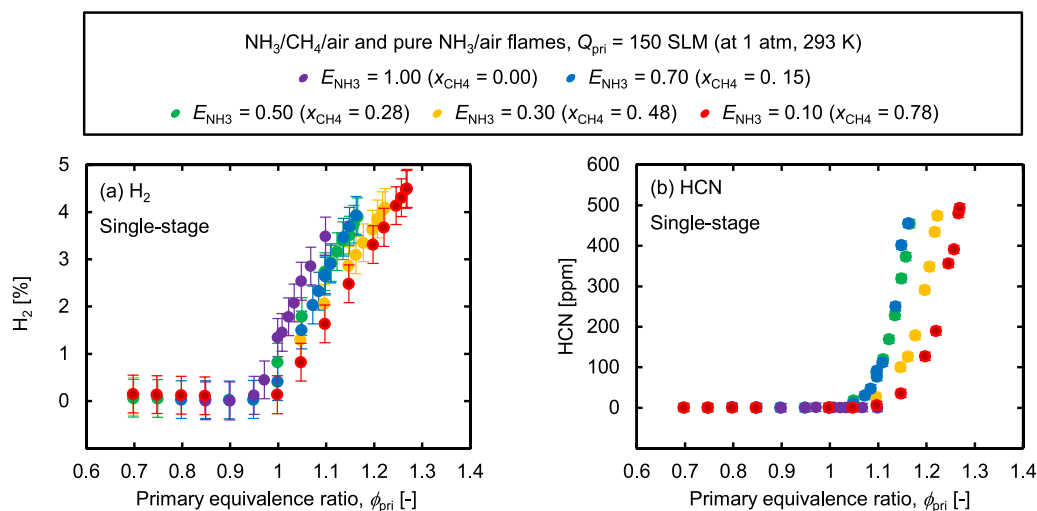


Fig. 6. (a) H_2 and (b) HCN emissions in the single-stage combustion of ammonia/methane/air and pure ammonia/air flames.

conditions may be that ϕ_{pri} with sufficient O and OH radicals can consume HCN shifted to the lean side with an increasing E_{NH_3} .

3.2.2. Ammonia/hydrogen/air flames

Fig. 7 shows the NO_x and unburned NH_3 emissions during the single-stage combustion of ammonia/hydrogen/air flames. The measurable equivalence ratios were limited by the N_2O (1000 ppm) and NO measurement ranges (5000 ppm) on the lean side, and the unburned NH_3 (5000 ppm) and H_2 (20%) measurement ranges on the rich side. Noise caused by thermoacoustic combustion instability was observed in ammonia/hydrogen/air flames but not in the ammonia/methane/air flames. The product gas measured from the ammonia/hydrogen/air

flames oscillated at smaller increments than those from the ammonia/methane/air flames. The oscillations caused periodic fluctuations of the product gas and prevented it from reaching a steady state for $\phi_{pri} = 0.70$ at $E_{NH_3} = 0.13$ ($x_{H_2} = 0.90$), where the noise was particularly loud. This makes it impossible to obtain the steady state values of the product gas. These conditions were excluded from the analysis because of the thermoacoustic instabilities.

The ϕ_{opt} for the simultaneous reduction of NO and unburned NH_3 was observed for all E_{NH_3} conditions in the ammonia/hydrogen/air flames, similar to that observed in the ammonia/methane/air and pure ammonia/air flames. The specific values of NO and unburned NH_3 at the ϕ_{opt} are discussed in Section 3.5. The contraction rate of the low-

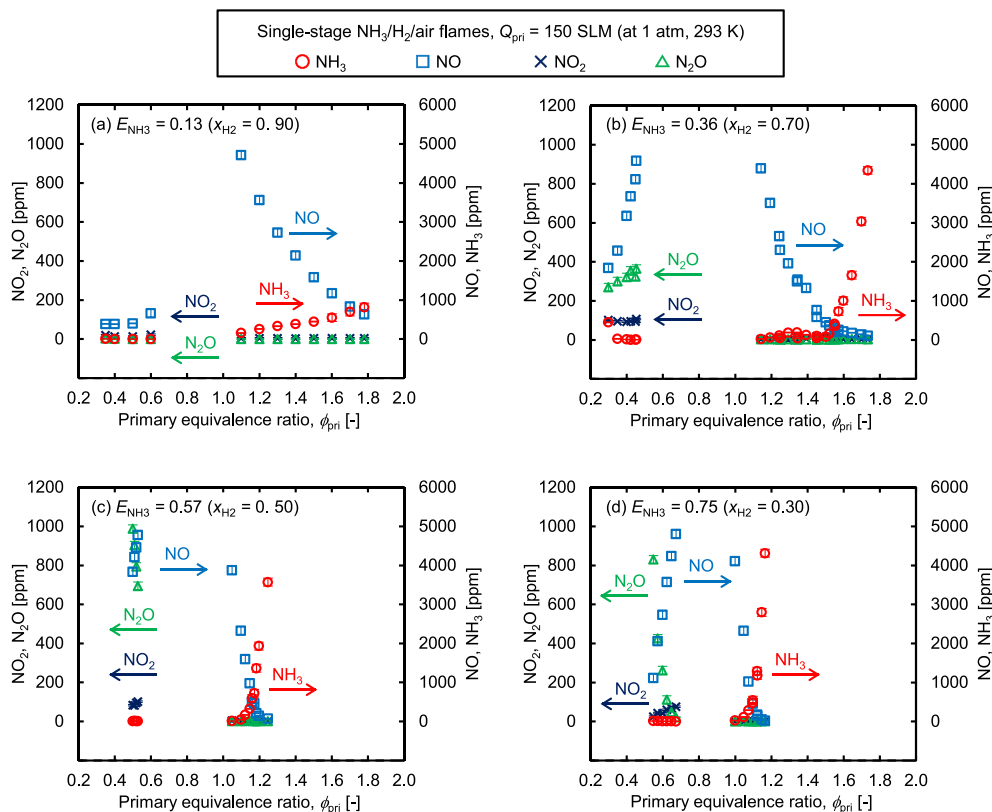


Fig. 7. NO_x and unburned NH_3 emissions in the single-stage combustion of ammonia/hydrogen/air flames at (a) $E_{NH_3} = 0.13$, (b) $E_{NH_3} = 0.36$, (c) $E_{NH_3} = 0.57$, and (d) $E_{NH_3} = 0.75$.

emission window with increasing E_{NH_3} was greater in the ammonia/hydrogen/air flames than in the ammonia/methane/air flames. NO peaked around $\phi_{\text{pri}} = 0.80$, which is leaner than the NO peaks for ammonia/methane/air and pure ammonia/air flames. Although the main component of NO_x under rich conditions was NO, some lean conditions emitted N_2O close to the measurement limit of 1000 ppm. Comparing the data in Figs. 5 and 7, the scales of the left vertical and horizontal axes of the two figures are different. This rapid increase in N_2O under lean conditions was reported by Mashruk et al. [48] and Hayakawa et al. [49]. Further, NO and N_2O under the lean conditions for $E_{\text{NH}_3} = 0.57$ and 0.75 showed a trade-off relationship, with the emissions remaining high even under their minimum conditions. As suggested by Hayakawa et al. [49], a low flame temperature under lean conditions suppresses the reduction of N_2O , which is highly temperature dependent. Further, the decrease in hydrogen-derived H radicals caused by an increase in E_{NH_3} was assumed to suppress the reduction of N_2O . The experimental conditions in this study, such as room temperature, atmospheric pressure, no air preheating, and the use of a quartz glass liner, which are different from those used in practical gas turbine combustors, may increase N_2O emissions under lean conditions. However, this phenomenon is noteworthy for the combustion. Measures to control N_2O emissions include elevating the ambient pressure and controlling the wall temperature of combustors [37]. Fig. 8 shows that H_2 is emitted stoichiometrically and increased linearly as the flame becomes richer. The emitted H_2 from the flame can include unburned H_2 in the fuel and H_2 produced by the ammonia oxidation under fuel-rich conditions. The H_2 emissions in $E_{\text{NH}_3} = 0.36$ ($x_{\text{H}_2} = 0.70$) were significantly lower than that under other conditions; this lower H_2 emissions were correlated with a slightly higher amount of unburned NH_3 under these conditions. This phenomenon was repeatedly observed in the setup used in this study.

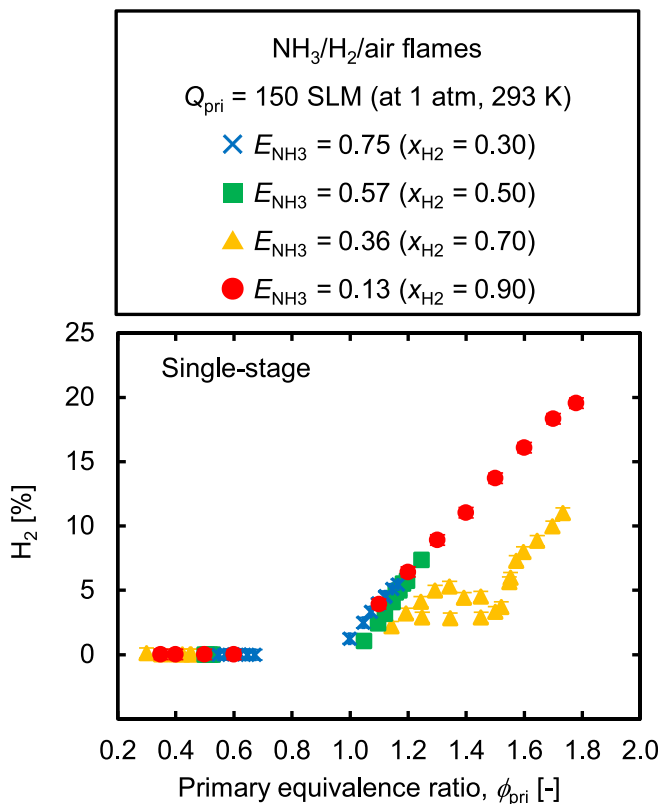


Fig. 8. H_2 emissions in the single-stage combustion of ammonia/hydrogen flames.

3.3. Two-stage combustion

3.3.1. Ammonia/methane/air flames

Fig. 9 shows the NO_x and unburned NH_3 emissions during the two-stage combustion of ammonia/methane/air and pure ammonia/air flames. The results for the other E_{NH_3} conditions are shown in Fig. S3. As shown in Fig. S4, the influence of ϕ_{ovt} on the product gas characteristics for two-stage combustion was small, as in the other combustors [38,39]. The unburned NH_3 , H_2 , and HCN were not released under all E_{NH_3} conditions, as shown in Figs. S6a and b. This is attributed to the unburned NH_3 , H_2 , and HCN from the primary combustion zone are completely burned in the secondary combustion zone. NO emissions exhibited a U-shaped trend with ϕ_{pri} . The minimum value of NO was observed at ϕ_{opt} for the two-stage combustion. The specific value of NO at ϕ_{opt} is discussed in Section 3.5. NO_2 exhibited the same characteristics as NO in the two-stage combustion. NO and NO_2 increase on the rich side of ϕ_{opt} because unburned NH_3 produced in the primary combustion zone is converted to NO_x in the secondary combustion zone. In other words, excessive unburned NH_3 from the primary combustion zone must be avoided in the rich-lean two-stage combustion to mitigate NO_x production in the secondary combustion zone. N_2O emissions were recorded at ϕ_{pri} where they were not emitted in the single-stage combustion and increased as ϕ_{pri} moved away from stoichiometry. This can be attributed to the low-temperature regions created by the secondary air at room temperature, which inhibited the reduction of N_2O .

3.3.2. Ammonia/hydrogen/air flames

Fig. 10 shows the NO_x and unburned NH_3 emissions during the two-stage combustion of ammonia/hydrogen/air flames. An enlarged version of the figure is shown in Fig. S5. Similar to the ammonia/methane/air and pure ammonia/air flames, ϕ_{opt} for the two-stage combustion realized the minimum value of NO. The specific value of NO at ϕ_{opt} is discussed in Section 3.5. The unburned NH_3 and H_2 from the primary combustion zone were completely burned in the secondary combustion zone, as shown in Fig. S7. The curvature of the U-shaped NO emissions trend was very subtle at $E_{\text{NH}_3} = 0.13$ ($x_{\text{H}_2} = 0.90$) and increased with an increase in E_{NH_3} . Compared with the ammonia/methane/air and pure ammonia/air flames, the low-emission window for the two-stage combustion was wider at lower E_{NH_3} and narrower at higher E_{NH_3} . Further, N_2O emissions were almost zero for all E_{NH_3} conditions in ammonia/hydrogen/air flames because the higher flame temperature and more H radicals compared with ammonia/methane/air flames may promote N_2O reduction.

3.4. NO conversion ratio of single- and two-stage combustion

The main purpose of the secondary air in rich-lean two-stage combustion is to burn unburned NH_3 and H_2 emitted from the primary combustion zone, but it also serves as a diluent for the product gas. To ensure a fair comparison of NO emissions from single- and two-stage combustion, the NO conversion ratio CR_{NO} was introduced in addition to the mole fraction. The CR_{NO} was defined by

$$CR_{\text{NO}} = \frac{X_{\text{NO}}}{X_{\text{NO, th}}}, \quad (7)$$

where X_{NO} and $X_{\text{NO, th}}$ represent the actual measured NO concentration and theoretically calculated NO concentration in the combustion products, assuming that all N atoms in ammonia are converted to Fuel-NO. The detailed calculation process is described in Section S3. The assumption in Eq. (7) is reasonable because the NO produced from ammonia flames is caused by Fuel-NO and not Thermal-NO [50]. In ammonia flames, the first rate-limiting reaction of the extended Zeldovich mechanism, $\text{N}_2 + \text{O} = \text{NO} + \text{N}$, acts as a reduction reaction and reduces NO [2].

Fig. 11 shows the CR_{NO} of the ammonia/methane/air, ammonia/

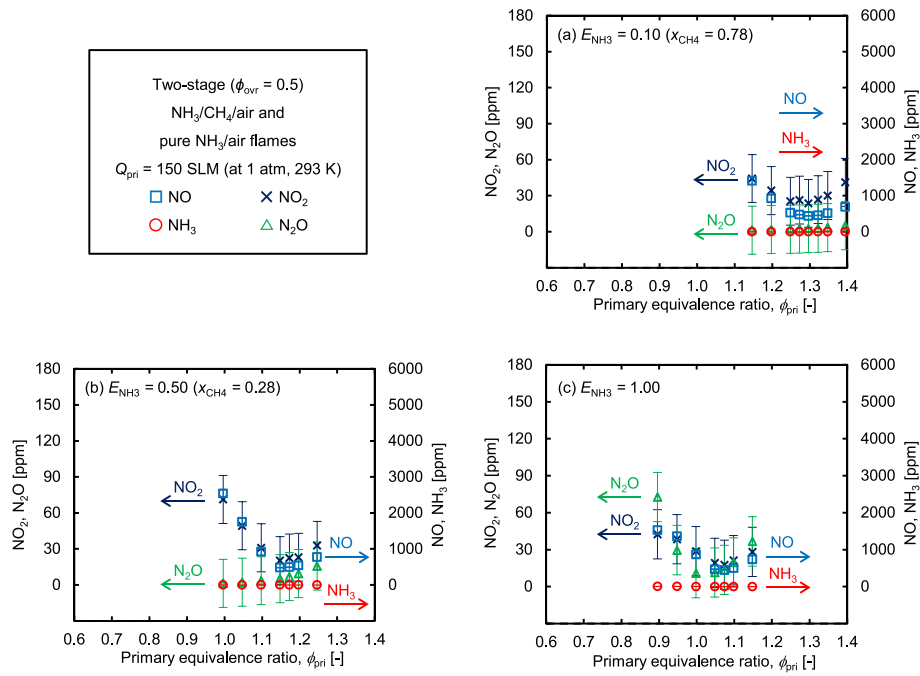


Fig. 9. NO_x and unburned NH_3 emissions in the two-stage combustion of ammonia/methane/air and pure ammonia/air flames at (a) $E_{\text{NH}_3} = 0.10$ and (b) $E_{\text{NH}_3} = 0.50$, and (c) $E_{\text{NH}_3} = 1.00$.

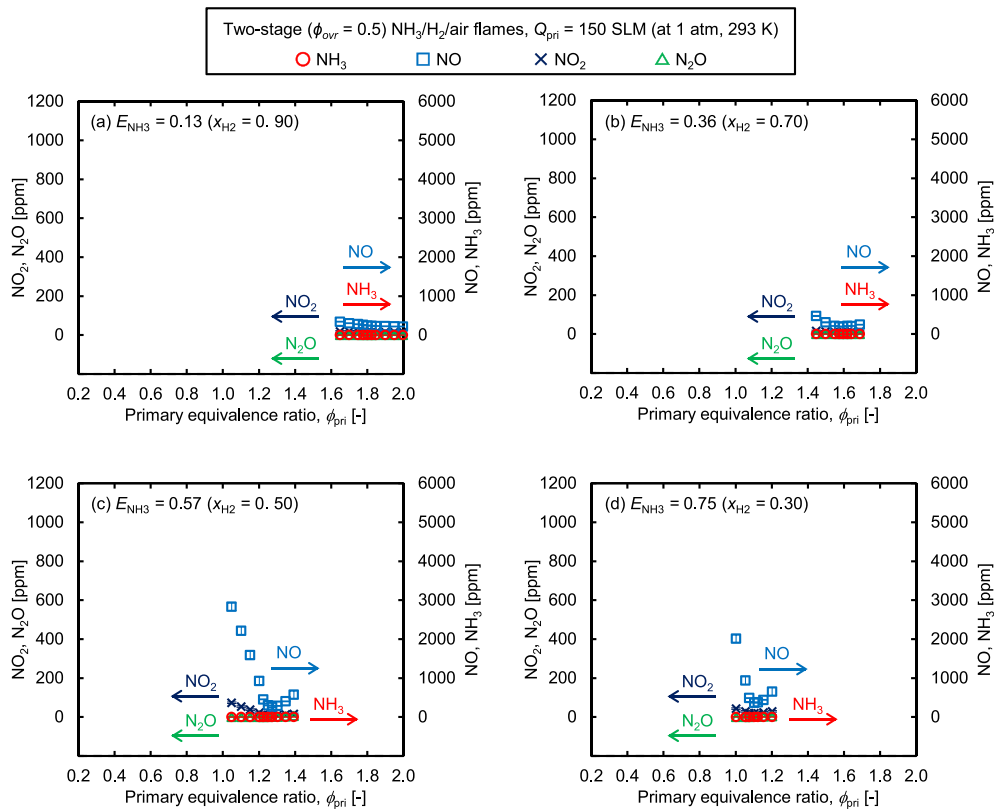


Fig. 10. NO_x and unburned NH_3 emissions in the two-stage combustion of ammonia/hydrogen/air flames at (a) $E_{\text{NH}_3} = 0.13$, (b) $E_{\text{NH}_3} = 0.36$, (c) $E_{\text{NH}_3} = 0.57$, and (d) $E_{\text{NH}_3} = 0.75$.

hydrogen/air, and pure ammonia/air flames in the combustion experiments. Under all conditions, CR_{NO} naturally had a good correlation with NO emissions. Comparing single- and two-stage combustion reveals that the CR_{NO} of the two-stage combustion was larger than that of the single-

stage combustion. This indicates that the unburned NH_3 emitted from the primary combustion zone was converted to NO in the secondary combustion zone of the two-stage combustion. There was no significant difference between the CR_{NO} of the single- and two-stage combustion at

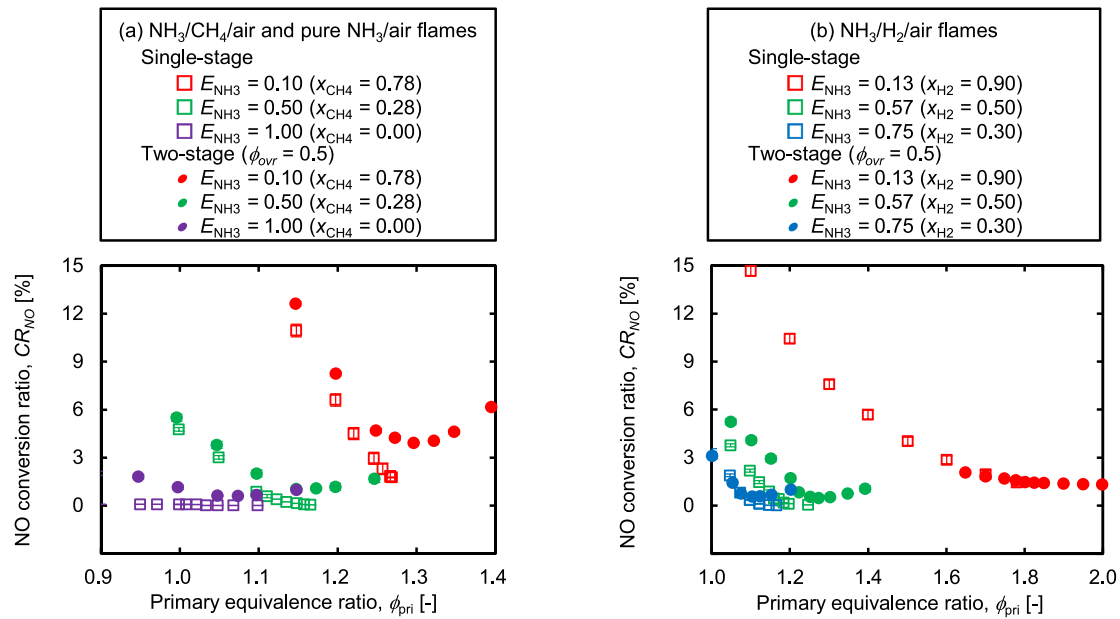


Fig. 11. NO conversion ratio CR_{NO} of (a) ammonia/methane/air and pure ammonia/air flames, and (b) ammonia/hydrogen/air flames.

$E_{NH_3} = 0.13$ ($x_{H_2} = 0.90$) for ammonia/hydrogen/air flames with the low unburned NH_3 , as shown in Fig. 7a. The difference in the CR_{NO} between the single- and two-stage combustion of ammonia/methane/air flames was greater than that of ammonia/hydrogen/air flames at each ϕ_{opt} because the unburned NH_3 emissions from the methane-blended fuel were larger than those from the hydrogen-blended fuel at a constant equivalence ratio. Further, this indicates that the HCN emitted from the primary combustion zone of the ammonia/methane/air flames was converted to NO in the secondary combustion zone, which increased the CR_{NO} . The larger difference in CR_{NO} between the single- and two-stage combustion of ammonia/hydrogen/air flames at $E_{NH_3} = 0.57$ under $\phi_{pri} < 1.2$ may result from higher levels of unburned NH_3 from the primary combustion zone in those conditions compared to other flames.

3.5. Optimum primary equivalence ratio for rich-lean two-stage combustion

All optimum primary equivalence ratios obtained experimentally and numerically in this study and the direct images of flames at ϕ_{opt} are listed in Table 2. The experimental ϕ_{opt} for the single-stage combustion was derived by the linear interpolation of each of the two plots of NO and unburned NH_3 close to the ϕ_{opt} shown in Figs. 5 and 7 and by finding the intersection of the two lines. The numerical ϕ_{opt} was also derived using numerical plots as well as the experimental ϕ_{opt} for the single-stage combustion. The experimental ϕ_{opt} for the two-stage combustion was derived by finding the minimum value of the function obtained by a fourth-order polynomial approximation of the NO plots shown in Figs. 9 and 10. The conditions for each direct image were different because the camera settings were adjusted to make the images similar to those of the human eye. In the direct image of the two-stage combustion, the ISO was amplified to observe the flame in the secondary combustion zone. In ammonia/methane/air flames, blue chemiluminescence from CH was observed in the primary combustion zone at low E_{NH_3} conditions, which transitioned to orange chemiluminescence from NH_2 ammonia α band and H_2O vapor with an increase in E_{NH_3} . Radiation spectrum analysis with a spectroscope (StellarNet, BLUE-Wave UVNb-50) was performed on a characteristic blue flame in the secondary combustion zone of the two-stage combustion of ammonia/methane/air flames; however, the signal was too weak to be measured. In ammonia/hydrogen/air flames, clear imaging was hindered by the red-hot base region of the glass liner attributed to the high flame temperature at low E_{NH_3} conditions;

however, orange chemiluminescence from NH_2 ammonia α band and H_2O vapor were observed at $E_{NH_3} = 0.75$ ($x_{H_2} = 0.30$).

Fig. 12a shows ϕ_{opt} for the single- and two-stage combustion of ammonia/methane/air, ammonia/hydrogen/air, and pure ammonia/air flames. The ϕ_{opt} for single- and two-stage combustion of ammonia/methane/air flames decreased linearly with an increase in E_{NH_3} . The ϕ_{opt} for single- and two-stage combustion of ammonia/hydrogen/air flames decreased exponentially with an increase in E_{NH_3} . For $E_{NH_3} > 0.60$, the ϕ_{opt} of ammonia/methane/air and ammonia/hydrogen/air flames were almost identical; however, for $E_{NH_3} < 0.60$, the difference widened rapidly. The numerical ϕ_{opt} was higher than the experimental ϕ_{opt} in each E_{NH_3} condition. Although the diffusion swirl combustor used in the study exhibited premixed-like characteristics because the fuel and air were well mixed at the bottom of the combustor, the combustion method differed from the numerical calculation of the preferable premixed flame. In addition, the experiments had a residence time. However, the qualitative trend of the exponential ϕ_{opt} with respect to E_{NH_3} was similar to that for numerical ϕ_{opt} . Hence, the factors that determine ϕ_{opt} can be discussed in terms of chemical reactions.

Comparing the experimental results for single- and two-stage combustion, the ϕ_{opt} for the two-stage combustion was slightly larger than that for the single-stage combustion under all conditions. For the primary combustion zone in two-stage combustion, a slightly richer mixture with less NO and more unburned NH_3 than ϕ_{opt} in single-stage combustion is suitable because NO emissions are determined by the sum of the NO produced in the primary and secondary combustion zones. Another possible reason is that the secondary air enhances the central recirculation zone and flows into the primary combustion zone, as discussed for the flow field in Section 3.1. The enhancement of recirculation led to an increase in the residence time, causing the experimental results to be closer to the numerical results. Further, the flow into the primary combustion zone dilutes ϕ_{pri} and apparently shifts ϕ_{opt} to the rich side. The large difference between the experimental and numerical results at $E_{NH_3} = 0.13$ ($x_{H_2} = 0.90$) are attributed to the fuel injection disturbing the recirculation and reducing the residence time. The volume flow rate of the fuel is larger than under other conditions because the ϕ_{opt} is large and the heating value of hydrogen is lower than that of ammonia and methane. Hence, there is a chance in that a high fuel injection flow rate will decrease ϕ_{opt} . The experimental result for two-stage combustion closer to the numerical result than that for the single-stage combustion at $E_{NH_3} = 0.13$ ($x_{H_2} = 0.90$) can be attributed to

Table 2

Optimum primary equivalence ratio ϕ_{opt} at various E_{NH_3} conditions of (a) ammonia/methane/air and pure ammonia/air flames and (b) ammonia/hydrogen/air flames. Note that the experiments and numerical simulations were performed on non-premixed and premixed flames, respectively.

(a) $\text{NH}_3/\text{CH}_4/\text{air}$ and pure NH_3/air flames										
E_{NH_3}	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.00
x_{CH_4}	0.78	0.61	0.48	0.37	0.28	0.21	0.14	0.09	0.04	0
#	1-1	1-2	1-3	1-4	1-5					
Single-stage ϕ_{opt} [-]	1.27	1.22	1.20	1.15	1.12	1.12	1.10	1.07	1.04	1.03
ISO [-]	400	320	400	400	100					
Shutter [1/s]	100	100	100	100	100					
#	2-1	2-2	2-3	2-4	2-5					
Two-stage ϕ_{opt} [-]	1.30	1.23	1.16	1.13	1.07					
ISO [-]	800	800	800	800	500					
Shutter [1/s]	100	100	100	100	100					
Sim. ϕ_{opt} [-]	1.43	1.38	1.35	1.31	1.27	1.24	1.21	1.17	1.15	1.12

(b) $\text{NH}_3/\text{H}_2/\text{air}$ flames										
E_{NH_3}	0.13	0.25	0.36	0.47	0.57	0.66	0.75	0.84	0.92	
x_{H_2}	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10	
#	1-1	1-2	1-3	1-4						
Single-stage ϕ_{opt} [-]	1.73	1.54	1.16	1.10						
ISO [-]	2500	2500	1000	100						
Shutter [1/s]	100	100	100	15						
#	2-1	2-2	2-3	2-4						
Two-stage ϕ_{opt} [-]	2.00	1.59	1.30	1.12						
ISO [-]	6400	3200	3200	800						
Shutter [1/s]	100	100	100	100						
Sim. ϕ_{opt} [-]	2.82	2.13	1.73	1.51	1.38	1.30	1.25	1.21	1.18	

the secondary air reducing the effect of large fuel injection.

Fig. 12b shows the minimum NO emissions at ϕ_{opt} of ammonia/methane/air, ammonia/hydrogen/air, and pure ammonia/air flames. Here, NO emissions were corrected at 16% O_2 to compare the single- and two-stage combustion. According to the method for obtaining ϕ_{opt} in this study, single-stage combustion emits the same amount of NH_3 as NO at ϕ_{opt} , whereas two-stage combustion ($\phi_{\text{ovr}} = 0.5$) emits no NH_3 because NH_3 is burned in the secondary combustion zone. In other words, the NO plots for single-stage combustion shown in Fig. 12b are synonymous with the NH_3 plots. The error bars for the three conditions for the ammonia/methane/air flames were larger than for the other conditions because these NO and unburned NH_3 before 16% O_2 correction exceeded 500 ppm and the FTIR gas analyzer changed its measurement range of NO and NH_3 from 500 ppm to 5000 ppm at 500 ppm.

In pure ammonia/air flames, the minimum NO and unburned NH_3 at ϕ_{opt} for single-stage combustion were ~ 90 ppm@16% O_2 . Two-stage combustion suppressed unburned NH_3 to almost zero but increased NO by ~ 80 ppm@16% O_2 compared to the single-stage combustion caused by unburned NH_3 emitted from the primary combustion zone converted to NO in the secondary combustion zone of the two-stage combustion. In ammonia/methane/air flames, the minimum NO and unburned NH_3 at ϕ_{opt} for single-stage combustion decreased with an increasing E_{NH_3} and were less than 100 ppm@16% O_2 at $E_{\text{NH}_3} < 0.70$.

Two-stage combustion suppressed unburned NH_3 to almost zero; however, it increased NO compared to the single-stage combustion and pure ammonia/air flames. This suggests that HCN and unburned NH_3 emitted from the primary combustion zone were oxidized to NO in the secondary combustion zone with excess air. In the ammonia/hydrogen/air flames, the minimum NO and unburned NH_3 at ϕ_{opt} for single-stage combustion were less than 80 ppm@16% O_2 and appear to decrease with increasing E_{NH_3} , although not as clearly as for ammonia/methane/air flames. The NO emissions at ϕ_{opt} for the two-stage combustion were equal to or lower than those for the single-stage combustion.

3.6. Effects of chemical kinetics

Nitrogen dilution experiments and numerical simulations at $\text{O}_2 = 17.5\%$ were performed on ammonia/methane/air flames to clarify the mechanism by which ϕ_{opt} decreases with increasing E_{NH_3} . The variation of ϕ_{opt} with E_{NH_3} is caused by a change in the reactivity of the mixture. Therefore, changes in the reactivity of the mixture caused by nitrogen dilution can help identify factors involved in the determination of ϕ_{opt} . The flow rates of air and nitrogen for the nitrogen dilution experiments were 125 and 25 SLM, respectively.

Fig. 13 shows the measured and simulated effects of nitrogen dilution on NO, unburned NH_3 , and OH in ammonia/methane/air flames at

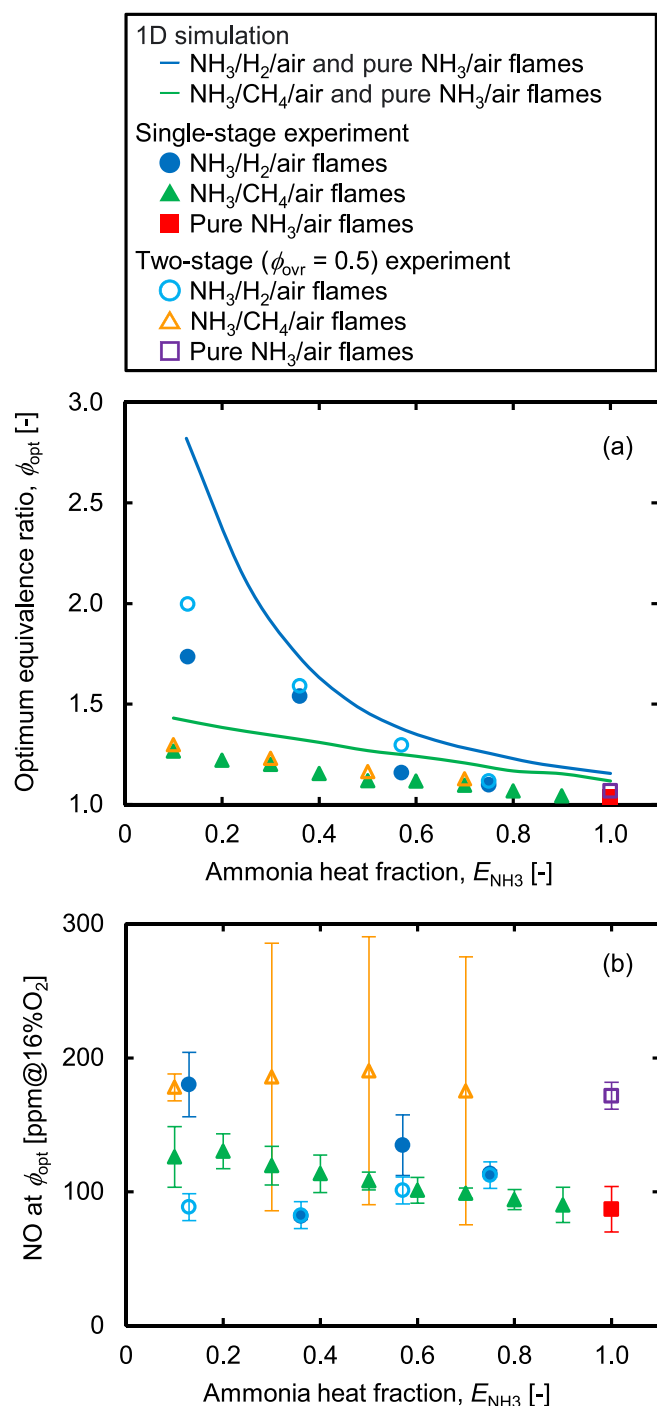


Fig. 12. (a) Optimum primary equivalence ratio ϕ_{opt} for the single- and two-stage combustion ($\phi_{ovr} = 0.5$) and (b) the NO emissions at ϕ_{opt} of ammonia/methane/air, ammonia/hydrogen/air, and pure ammonia/air flames.

$E_{NH_3} = 0.50$ ($x_{CH_4} = 0.28$). Fig. 13b represents the numerical mole fractions of NO, NH₃, and OH at 40 mm downstream from the point of maximum heat release rate. The ϕ_{opt} for the simultaneous reduction of NO and unburned NH₃ was also observed in the nitrogen dilution experiments and numerical simulations. Nitrogen dilution reduced the maximum value of NO, as shown by Khateeb et al. [21]. The figures show that the NO emissions decreased while unburned NH₃ emissions increased with nitrogen dilution, thereby shifting ϕ_{opt} towards the lean side. These trends are more evident in the numerical results shown in Fig. 13b. In other words, nitrogen dilution decreased ϕ_{opt} . The numerical results showed that the mole fraction of OH decreased at all equivalent

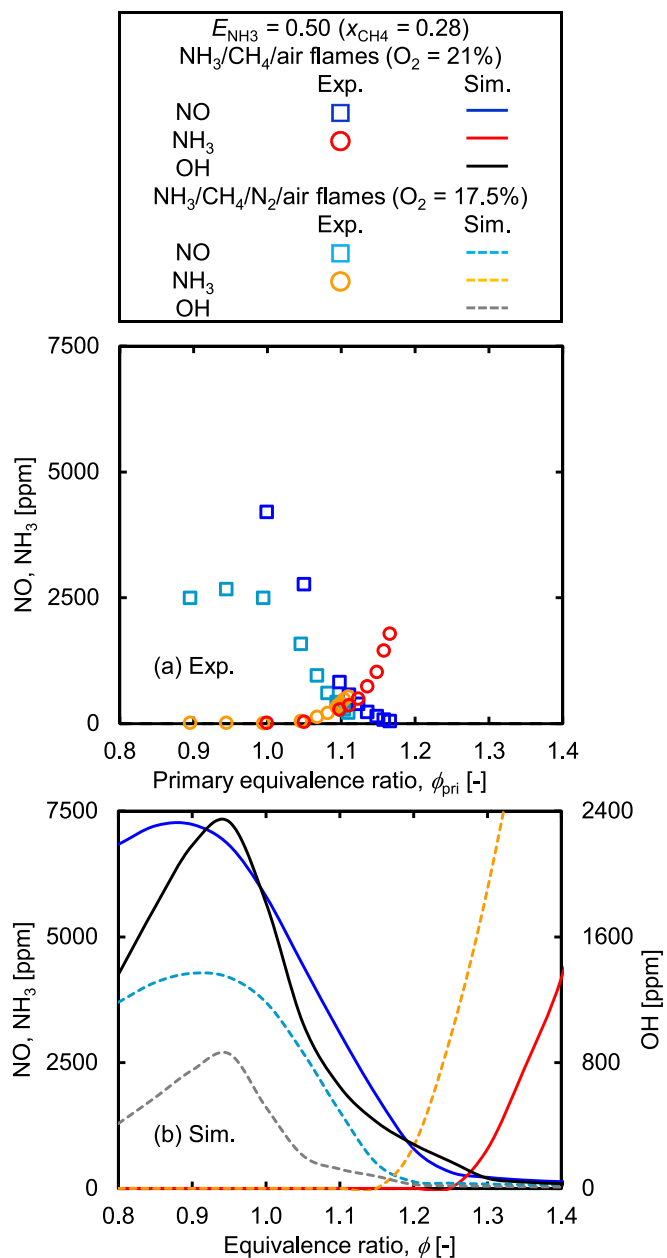


Fig. 13. (a) Experimental and (b) numerical results of the effect of nitrogen dilution on NO, unburned NH₃, and OH of ammonia/methane/air flames at $E_{NH_3} = 0.50$.

ratios. OH radicals, which are important for NO formation and NH₃ consumption, can contribute to the observed trends in the NO and NH₃ emissions profiles, and consequently, the shift in ϕ_{opt} . In ammonia/air flames, OH radicals drive the largest heat release reaction: $NH_3 + OH = NH_2 + H_2O$. O and OH radicals were dominant in the formation of NO from NH₂ and NH via HNO, and the role of OH radicals was important on the rich side because of the lack of an oxidizer. Therefore, NO emissions from ammonia/air flames continue to decrease, whereas the unburned NH₃ emissions start to increase when a certain amount of OH radicals is no longer supplied. The suppression of OH radicals by nitrogen dilution can shift ϕ_{opt} towards the lean side.

Fig. 14 shows the variation of OH mole fractions with E_{NH_3} at $\phi = 0.8, 1.0$, and 1.2 of ammonia/methane/air and ammonia/hydrogen/air flames. Note that these OH mole fractions were obtained by numerical simulation. The OH mole fraction of ammonia/methane/air flames decreased linearly with an increasing E_{NH_3} . In contrast, the OH mole

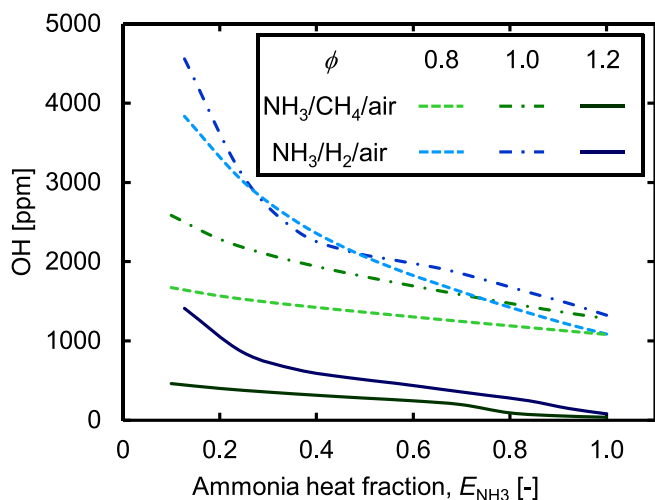


Fig. 14. Variation of the OH mole fraction with respect to E_{NH_3} at $\phi = 0.8, 1.0$, and 1.2 of ammonia/methane/air and ammonia/hydrogen/air flames.

fraction of the ammonia/hydrogen/air flames decreased exponentially with an increasing E_{NH_3} . These trends are consistent with the variation of ϕ_{opt} with E_{NH_3} shown in Fig. 11a, strongly suggesting that OH radicals are related to the determination of ϕ_{opt} . Fig. S8 shows that the rate of decrease of OH mole fraction under rich conditions became smaller with an increasing ϕ for all E_{NH_3} conditions. Under high E_{NH_3} conditions, OH changes more rapidly with a small change in the equivalence ratio because the ϕ_{opt} is closer to stoichiometry than that under low E_{NH_3} conditions. This provides a basis for narrowing the low-emission window with increasing E_{NH_3} shown in Figs. 5, 7, 9, and 10.

The contribution of ammonia/methane/air and ammonia/hydrogen/air flames to the formation of OH radicals in reactions with high sensitivity to OH was investigated for each flame to clarify how OH radicals are formed. Fig. 15 shows the integrated ROP of OH $I_{\text{R},i}$ for the top five reactions with the highest sensitivity to OH in the stoichiometric ammonia/methane/air and ammonia/hydrogen/air flames. The horizontal axes in Fig. 15a and b differ by a factor of 10. In both the ammonia/methane/air and ammonia/hydrogen/air flames, OH was produced through (R1) $\text{H} + \text{O}_2 = \text{O} + \text{OH}$, (R2) $\text{O} + \text{H}_2 = \text{H} + \text{OH}$, and (R3) $\text{O} + \text{H}_2\text{O} = \text{OH} + \text{OH}$, and consumed through (R4) $\text{OH} + \text{H}_2 = \text{H} + \text{H}_2\text{O}$ and (R5) $\text{OH} + \text{NH}_3 = \text{NH}_2 + \text{H}_2\text{O}$. In ammonia/methane/air flames, (R6) $\text{OH} + \text{CH}_4 = \text{CH}_3 + \text{H}_2\text{O}$ also contributed significantly to the consumption reaction. This finding indicated that H and H_2 are important to produce OH. In pure ammonia/air flames, $\text{NH}_x + \text{H}$ plays a key role in H_2 regeneration, and H_2 can be formed by the reactions of N_2H_2 and HNO with H [18]. In ammonia/methane/air flames, H_2 is produced by the reactions of CH_4 and CH_2O with H [18]. In the ammonia/hydrogen/air flames, H radicals increased with a decrease in E_{NH_3} , and the variation trend with E_{NH_3} was similar to that of the laminar burning velocity [51]. The integrated ROP of OH for (R1), (R2), (R3), and (R4) decreased linearly with an increasing E_{NH_3} for the ammonia/methane/air flames and exponentially with increasing E_{NH_3} for the ammonia/hydrogen/air flames. Further, the sum of the integrated ROP of OH for all reactions showed the same trend as the integrated ROP of OH for (R1), (R2), (R3), and (R4), suggesting that (R1), (R2), (R3), and (R4) significantly affect the variation of the OH mole fraction with respect to E_{NH_3} shown in Fig. 14 and contribute to the determination of ϕ_{opt} .

4. Conclusions

The optimum primary equivalence ratios ϕ_{opt} for the simultaneous reduction of NO and unburned NH_3 in ammonia/methane/air and ammonia/hydrogen/air flames were determined experimentally and

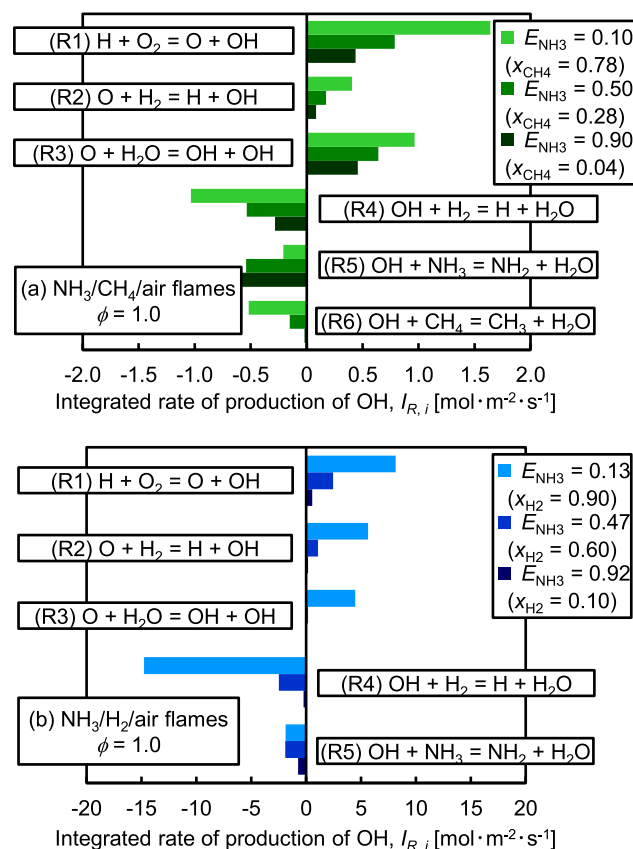


Fig. 15. Integrated rate of production (ROP) of OH in stoichiometric (a) ammonia/methane/air and (b) ammonia/hydrogen/air flames.

numerically over a wide range of ammonia heat fractions ($E_{\text{NH}_3} = 0.10$ – 1.00). The effect of the simultaneous reduction of NO and unburned NH_3 in the rich-lean two-stage combustion was examined under atmospheric pressure. The dominant factor in the determination of ϕ_{opt} was explored using 1D flame numerical simulations in terms of chemical kinetics. The major findings of this study are summarized as follows.

1. The ϕ_{opt} for the single- and two-stage combustion of ammonia/methane/air flames decreases with increasing E_{NH_3} .
2. The ϕ_{opt} for the single- and two-stage combustion of ammonia/hydrogen/air flames decreases with increasing E_{NH_3} more rapidly than that of ammonia/methane/air flames.
3. An increase in E_{NH_3} narrows the low-emission window and increases the sensitivity of the equivalence ratio to NO and unburned NH_3 near ϕ_{opt} . This phenomenon is more pronounced in the ammonia/hydrogen/air flames than in the ammonia/methane/air flames.
4. The ammonia mixing ratio changes the emission characteristics in the primary combustion zone of H_2 and HCN for ammonia/methane/air flames and H_2 for ammonia/hydrogen/air flames.
5. The ϕ_{opt} for the two-stage combustion of ammonia/hydrogen/air flames is equal to or lower than that for the single-stage combustion, whereas the ϕ_{opt} for the two-stage combustion of ammonia/methane/air or pure ammonia/air flames is larger than that for the single-stage combustion.
6. OH radical is one of the dominant factors for determining the ϕ_{opt} in the rich-lean two-stage combustion of ammonia flames.

CRediT authorship contribution statement

Hirofumi Yamashita: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Akihiro Hayakawa: Writing – review & editing, Supervision, Funding acquisition. **Ekenechukwu C. Okafor:** Writing – review & editing. **Sophie Colson:** Writing – review & editing. **K. D. Kunkuma A. Somarathne:** Writing – review & editing. **Taku Tsujimura:** Writing – review & editing, Supervision. **Shintaro Ito:** Writing – review & editing, Resources. **Masahiro Uchida:** Writing – review & editing, Supervision, Resources. **Taku Kudo:** Writing – review & editing, Supervision. **Hideaki Kobayashi:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2024.131598>.

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